Experiment -2

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Design a UI where users recall visual elements (e.g., icons or text chunks). Evaluate the effect of

chunking on user memory.

FRAME 1:

Chunking Analysis of the Cosmic Space UI (Memory Recall Task)

Chunking is a cognitive technique that organizes information into smaller, meaningful groups, helping users process and recall visual elements more effectively. The **Cosmic Space UI** applies chunking principles through layout, visual hierarchy, and minimal interaction elements to support user memory and focus.

1. Purpose of the Screen

The purpose of this screen is to introduce the task to users and explain how the memory recall activity works. It prepares users mentally by setting clear expectations before interaction begins.

2. Key Elements & UI Components

- **Title Text**: Clearly states the task and grabs attention.
- **Instructional Content**: Step-by-step guidance presented in a structured manner.
- **Visual Design Elements**: Minimal visuals that support understanding without distraction.
- **Primary Action Button (Start/Play)**: Indicates the next step in the flow.

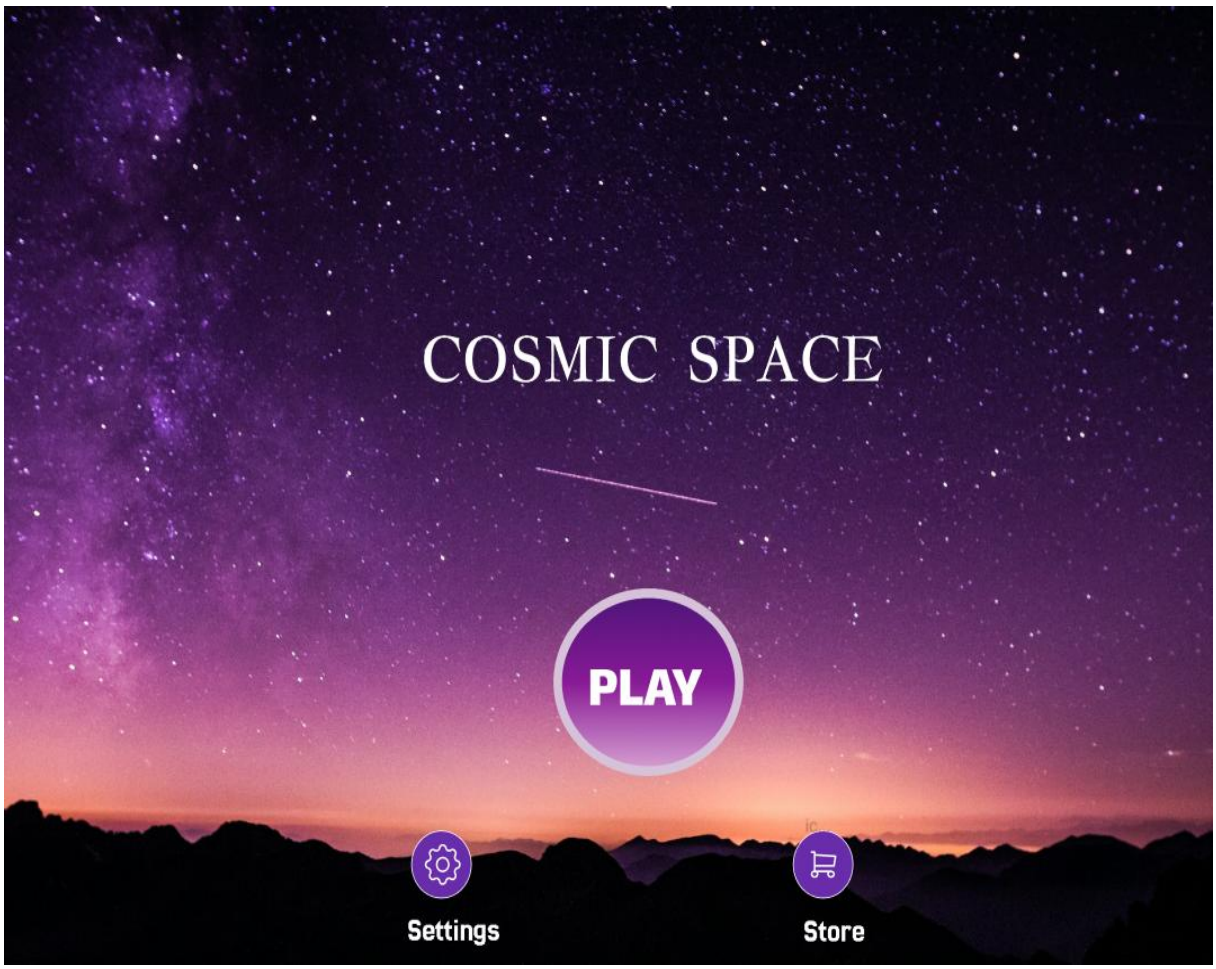
3. How This Phase Works

- Users read the instructions presented on the screen.
- Information is delivered in small, sequential chunks to aid comprehension.

- Users understand the task rules, time limits, and goals before proceeding.

4. Cognitive & UX Benefits

- **Chunking** reduces information overload by breaking instructions into manageable steps.
- **Improved comprehension** through clear hierarchy and spacing.
- **Lower cognitive effort** by using simple language and structured layout.
- **Better task readiness** as users know what to expect.



FRAME 2: Chunking Analysis of the "How to Play" Screen

The "How to Play" screen applies chunking principles to clearly communicate gameplay actions while supporting user memory and recall. By organizing

instructions into visual and functional groups, the interface reduces cognitive load and improves learning efficiency.

1. Purpose of the Screen

The purpose of this screen is to visually demonstrate core gameplay actions in a simple and intuitive manner. It bridges the gap between written instructions and actual gameplay.

2. Key Elements & UI Components

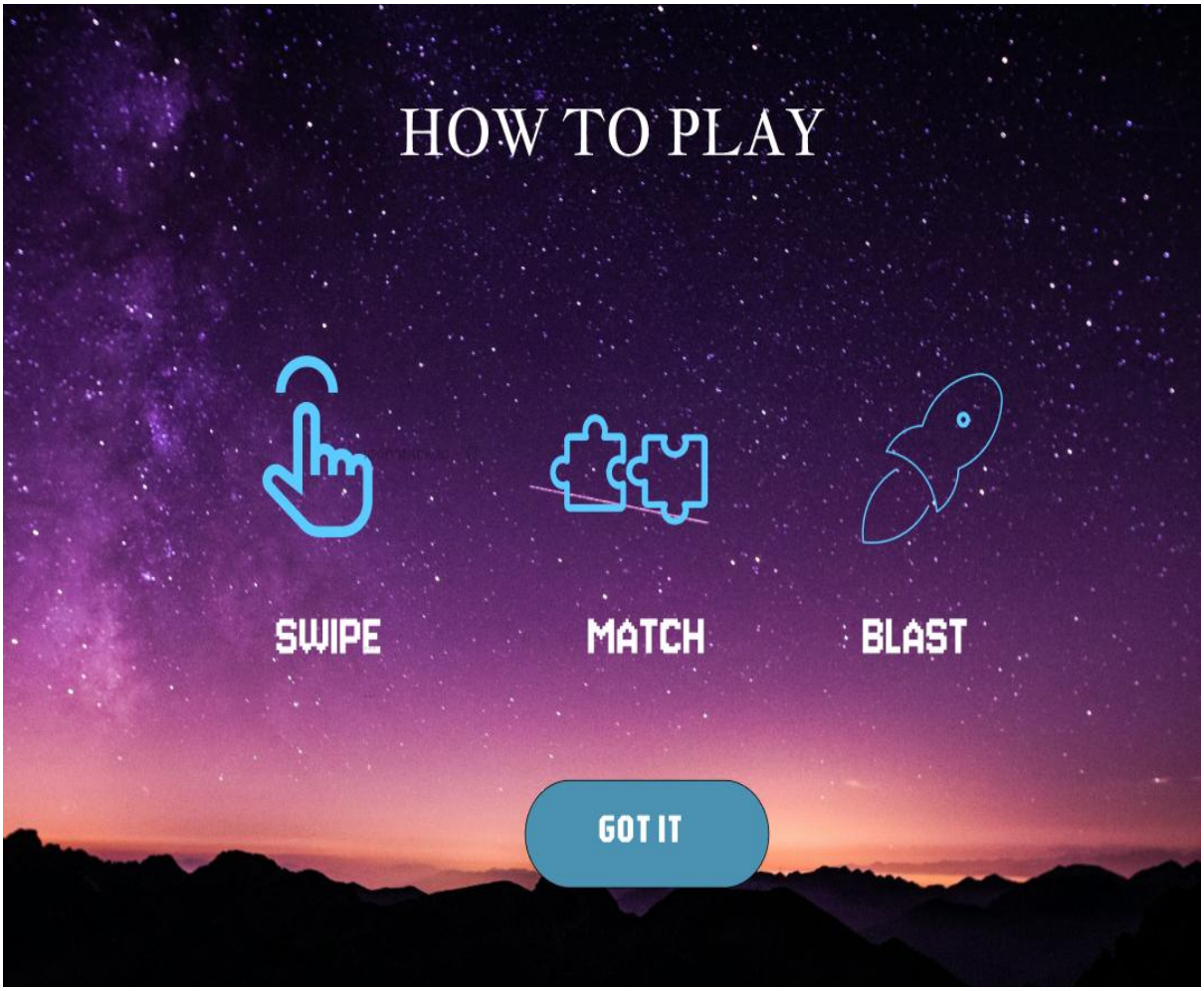
- Action Icons: Swipe, Match, and Blast symbols.
- Short Text Labels: Single-word descriptions under each icon.
- Title (“How to Play”): Provides clear context.
- Confirmation Button (“Got It”): Allows users to proceed once they understand.

3. How This Phase Works

- Users observe the three main actions represented visually.
- Each icon-text pair communicates a single interaction method.
- Users confirm understanding and move forward.

4. Cognitive & UX Benefits

- Dual coding (icon + text) strengthens memory retention.
- Action chunking helps users remember steps during gameplay.
- Reduced reading load supports faster learning.
- Higher recall accuracy when actions are needed later.



FRAME 3: Chunking Analysis of the “Select Level” Screen

The “Select Level” screen applies chunking principles to support decision-making and memory recall by organizing options into clear, structured units. The interface minimizes cognitive effort while guiding users through progressive task selection.

1. Purpose of the Screen

The purpose of this screen is to allow users to choose the difficulty or progression level before gameplay begins. It supports structured decision-making.

2. Key Elements & UI Components

- Screen Title (“Select Level”): Clarifies the task.

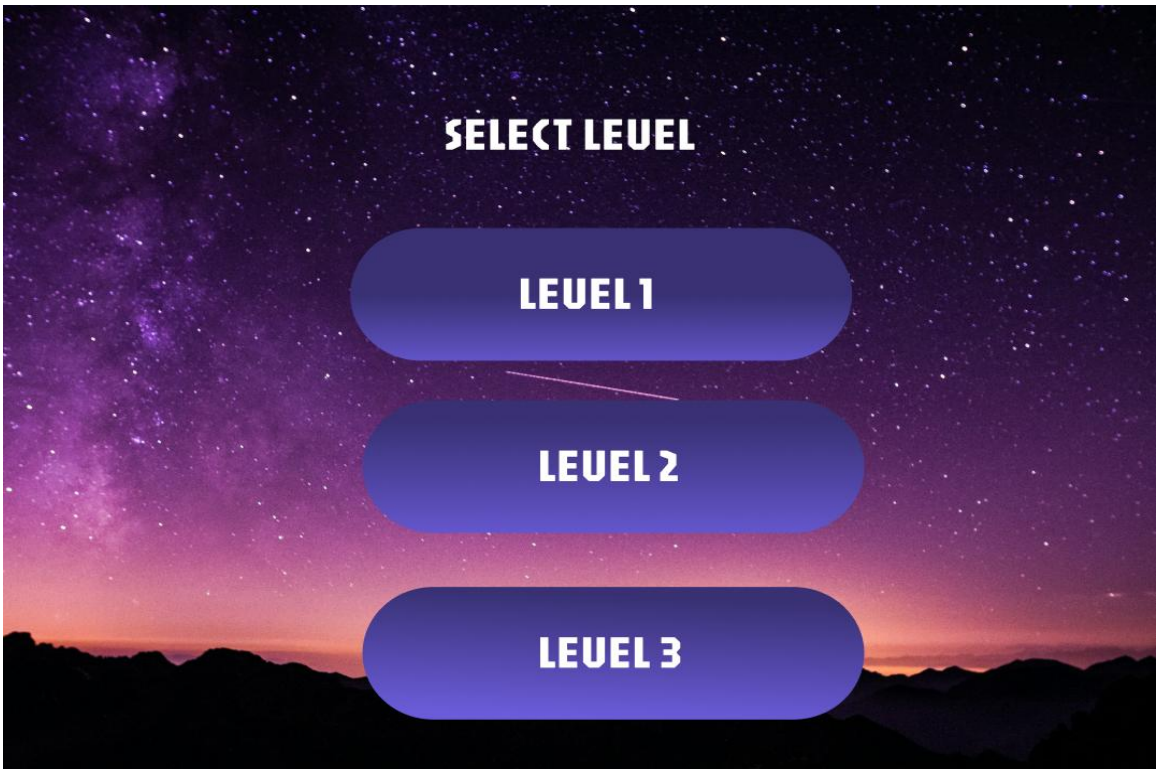
- Level Buttons (Level 1, Level 2, Level 3): Clearly labeled selectable options.
- Consistent Button Design: Uniform shape, size, and color.

3. How This Phase Works

- Users view all available levels at once.
- Levels are arranged vertically, suggesting progression.
- Users select one option to begin gameplay.

4. Cognitive & UX Benefits

- Limited choices reduce decision fatigue.
- Visual consistency improves recognition and recall.
- Clear hierarchy speeds up selection.
- Reduced cognitive load through simple layout.



FRAME 4: Chunking Analysis of the Gameplay Screen

The gameplay screen applies chunking principles to help users process multiple visual elements simultaneously while maintaining focus and memory recall. By organizing information into functional and spatial groups, the interface reduces cognitive overload during active interaction.

1. Purpose of the Screen

The purpose of this screen is to enable users to actively engage in the core gameplay task, where they interact with visual elements to match and recall information. It serves as the main interaction phase, combining decision-making, memory recall, and real-time feedback.

2. Key Elements & UI Components

- Score Display: Shows the user's current score, providing performance feedback.
- Level Indicator: Displays the current level to maintain context and progression awareness.
- Fuel Bar: Acts as a time/resource constraint guiding user urgency.
- Pause and Exit Buttons: Allow users to control gameplay flow.
- Hexagonal Tiles: The primary interactive elements containing symbols (H, Li, O, Xe).
- Consistent Background Theme: Maintains visual continuity without distracting from gameplay.

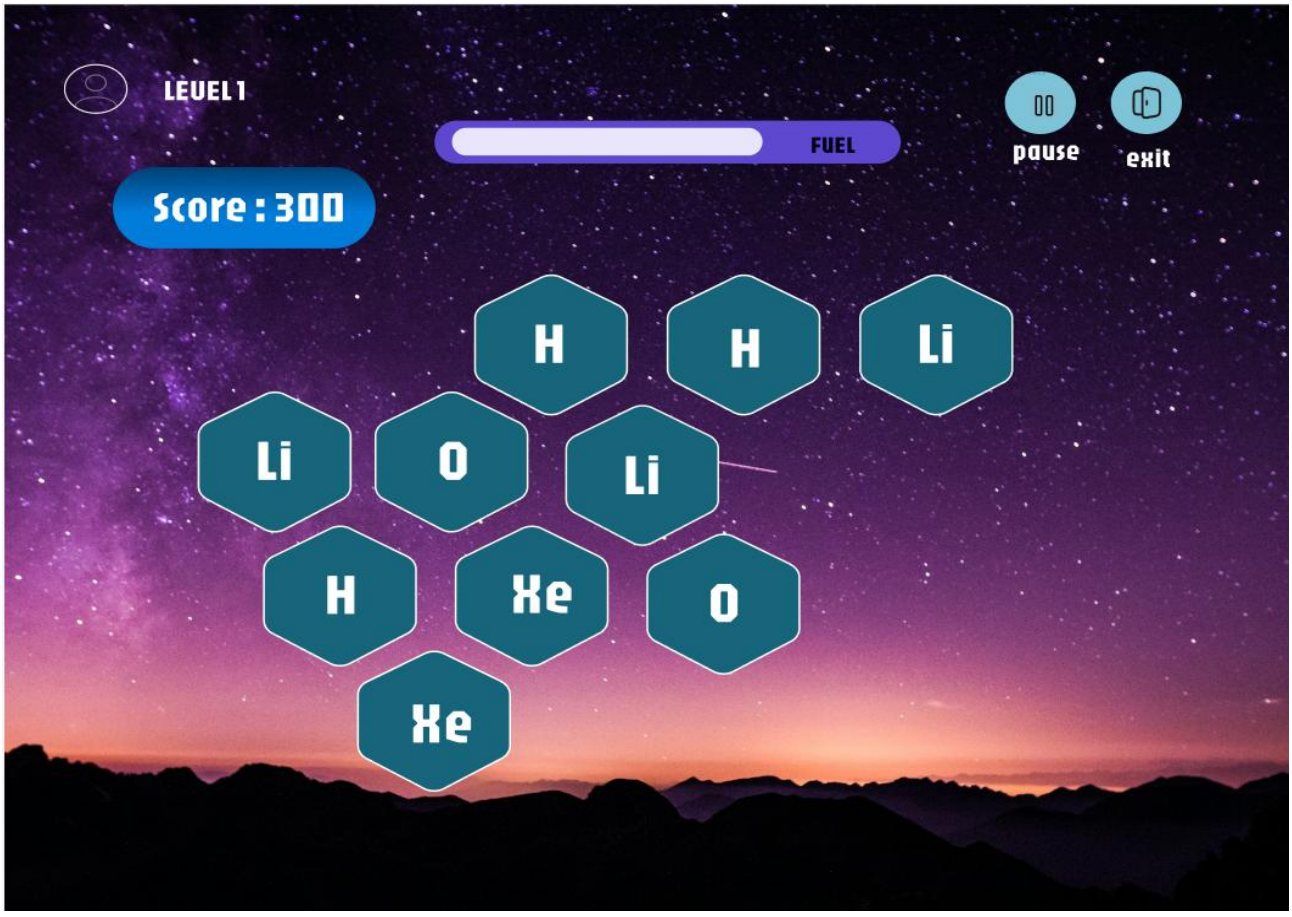
3. How This Phase Works

- Users interact with the hexagonal tiles to identify, match, or recall correct elements.
- The symbols require users to rely on short-term memory and pattern recognition.
- The fuel bar decreases over time, encouraging efficient decision-making.
- Scores update based on correct actions, reinforcing learning through feedback.

- Users can pause or exit without disrupting their understanding of the game state.

4. Cognitive & UX Benefits

- Chunking: Tiles with identical shapes and colors are perceived as one group, reducing visual complexity.
- Improved Recall: Repeated exposure to symbols strengthens memory encoding.
- Reduced Cognitive Load: Functional separation of controls, status, and gameplay elements improves clarity.
- Faster Recognition: Short symbols and consistent layouts support recognition over recall.
- Enhanced Focus: Central placement of interactive elements keeps user attention on the primary task.



FRAME 5: Result / Feedback Screen Analysis

performance information is grouped into a single results card, combining level status, star rating, and time taken. This chunked presentation allows users to quickly understand outcomes without processing scattered details. The clear separation of feedback and action buttons reduces cognitive load and supports faster decision-making.

1. Purpose of the Screen

The purpose of this screen is to provide users with clear feedback on their performance after completing a level. It marks the end of a gameplay cycle and helps users reflect on their results before deciding the next action.

2. Key Elements & UI Components

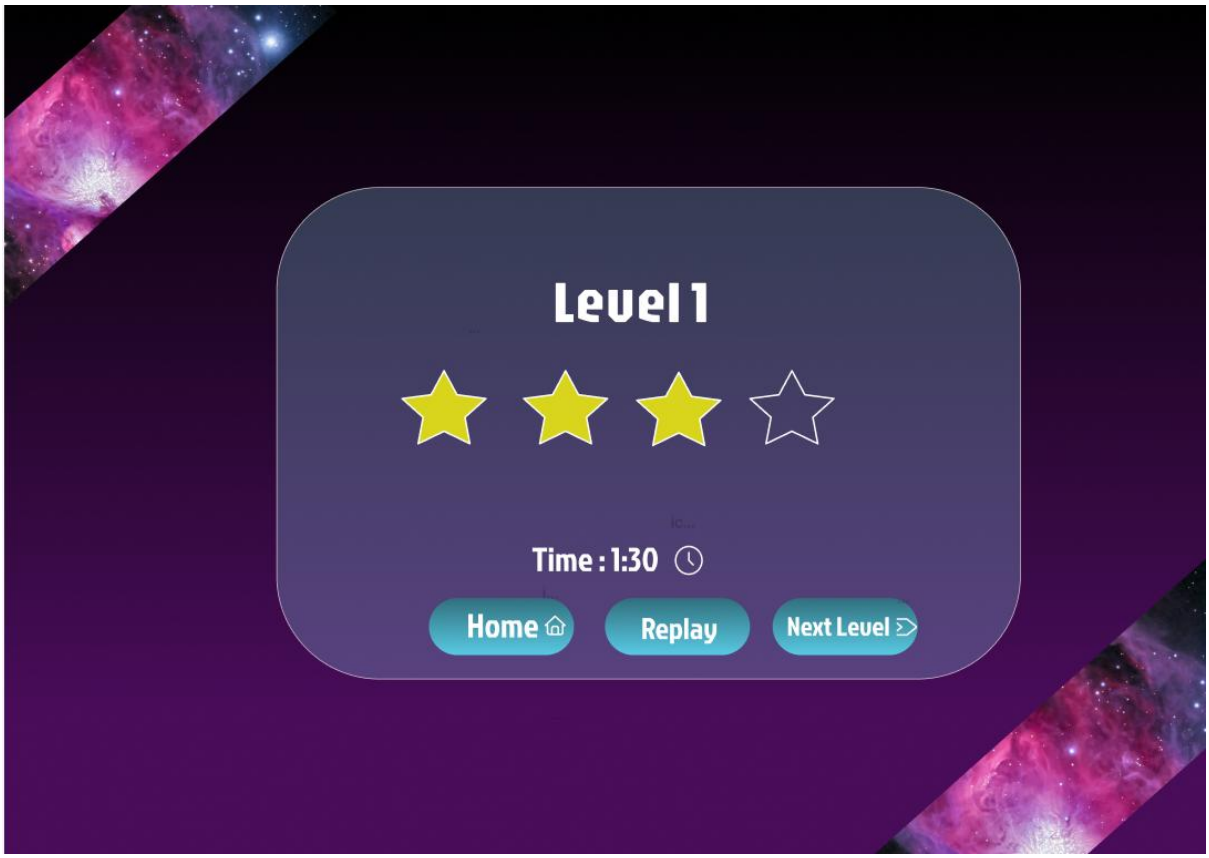
- **Level Indicator (Level 1)**: Confirms which level was completed.
- **Star Rating System**: Visually represents performance and achievement.
- **Time Taken Display**: Shows how long the user took to complete the level.
- **Action Buttons**:
    - **Home** – Return to the main menu
    - **Replay** – Retry the level
    - **Next Level** – Progress forward
- **Central Result Card**: Groups all feedback elements into a single focal area.

3. How This Phase Works

- After completing the level, users are shown their performance results.
- Star ratings instantly communicate success without requiring detailed analysis.
- The time metric allows users to compare efficiency across attempts.
- Users choose their next step (exit, retry, or continue), closing the feedback loop.

4. Cognitive & UX Benefits

- **Chunked Feedback**: Performance information is grouped into one central card, reducing visual scanning effort.
- **Visual Rewards**: Star ratings provide instant positive reinforcement and motivation.
- **Recognition Over Recall**: Clear icons and labels help users decide next actions quickly.
- **Motivation & Learning**: Immediate feedback supports reflection and encourages improvement.
- **Low Cognitive Load**: Limited choices prevent decision fatigue at the end of gameplay.



Prototype Link:

https://www.figma.com/proto/8A2NDS3opipDgEJw4Er4Em/Untitled?node-id=10-78&p=f&t=NRJc9k5Ur4gXrIn5-1&scaling=min-zoom&content-scaling=fixed&page-id=0%3A1